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66. When 1-pentyne (A) is treated with 4N alcoholic KOH at 175°C, it is converted slowly into an equilibrium mixture of 1.3% 1-pentyne(A) , 95.2% 2-pentyne(B) and 3.5% of 1,2 pentadiene (C) the equilibrium was maintained at 175°C, calculate ΔG^0 for the following equilibria.

$B \rightleftharpoons A \qquad \Delta G_1^0 = ?$

$B \rightleftharpoons C \qquad \Delta G_2^0 = ?$

67. At 33K, N_2O_4 is fifty percent dissociated. Calculate the standard free energy change at this temperature and at one atmosphere.

68. The standard enthalpies of formation of SO_2 and SO_3 are –297 kJ mol^{-1} and –396 kJ mol^{-1} respectively. Calculate the standard enthalpy of reaction for the reaction: $SO_2 + \frac{1}{2} O_2 \rightarrow SO_3$

69. For the reaction at 298 K : 2A +B → C

ΔH=400 J mol^{-1} ; ΔS = 0.2 JK^{-1} mol^{-1} Determine the temperature at which the reaction would be spontaneous.

70. Find out the value of equilibrium constant for the following reaction at 298K, 2 $NH_3(g) + CO_2(g) \rightleftharpoons NH_2CONH_2(aq) + H_2O(l)$ Standard Gibbs energy change, ΔG_r^0 at the given temperature is –13.6 kJ mol–1.

71. A gas mixture of 3.67 lit of ethylene and methane on complete combustion at 25°C and at 1 atm pressure produce 6.11 lit of carbondioxide. Find out the amount of heat evolved in kJ, during this combustion. ($\Delta H_C(CH_4)$= – 890 kJ mol^{-1} and ($\Delta H_C(C_2H_4)$= –1423 kJ mol^{-1}